

$$Y_e = (a + \bar{I}) / (1 - b) \quad \dots (2.5)$$

We depict equilibrium level of output in Fig. 2.3. Along the horizontal axis we measure output or income (recall that income is generated in the process of production of output, so we use the terms income and output interchangeably). Along the vertical axis we measure aggregate demand and its components (C and I).

Note that OL is a line which makes an angle of 45° with the horizontal axis, so that a perpendicular from this line such as AY_1 , forms an isosceles triangle AOY_1 , with $\angle OAY_1 = \angle AOY_1 = 45^\circ$, and as per the property of isosceles triangles, the two sides OY_1 and AY_1 are equal. Using this construct we are able to compare vertical distances (demand) with the horizontal distances (output or income). Note that at A , where the consumption function cuts OL , consumption expenditure AY_1 is exactly equal to income OY_1 , so that saving is zero.

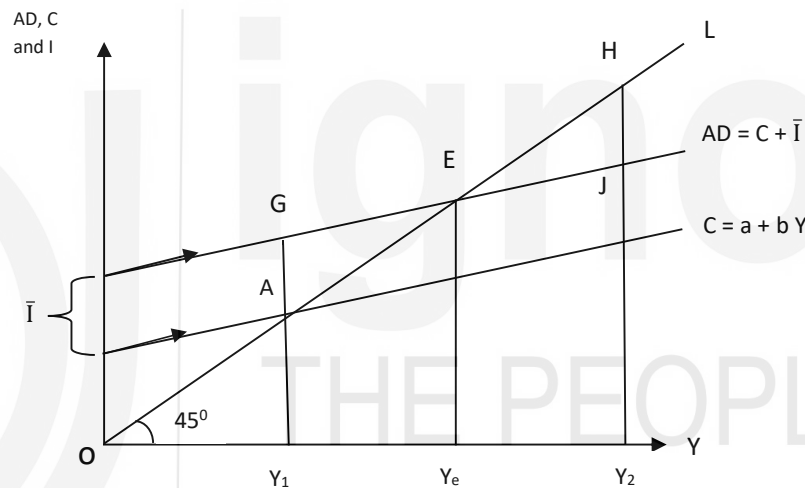


Fig. 2.3: Equilibrium Output in a Two-Sector Economy

In Fig. 2.3 we derive the aggregate demand curve (AD) by adding the investment ($\bar{I}$) and consumption (C) functions. Since investment is autonomous the line AD is parallel to the consumption function, i.e., slope of AD is the same as the slope of the consumption function. Equilibrium output is attained at E , at the intersection of AD with OL where aggregate demand EY_e is equal to aggregate output OY_e . Note that the equilibrium is unique, as there is only one level of income OY_e at which aggregate demand is equal to aggregate supply. In this case firms will be able to sell exactly as much as they had planned and hence there will be no unplanned changes in their stock of inventories.

For all income levels less than OY_e , i.e., to the left of E , there is excess demand for output. For example, when aggregate income is $OY_1 (=AY_1)$, the corresponding level of aggregate demand GY_1 is greater than the supply of output $AY_1 (=OY_1)$. In this case since demand is more than the output produced by firms, they will have to run down their stock of inventories. So in the next

period, firms will increase production and output would move toward OY_e . This process of increase in output (and hence income) will continue till OY_e is reached, where the amount produced (OY_e) being exactly equal to the amount demanded (EY_e), there is no unplanned changes in the stock of inventories and hence no further change in production levels.

Similarly, to the right of E (i.e., at all income levels greater than OY_e), there will be excess supply of output. When income is OY_2 , the corresponding level of aggregate demand JY_2 is less than the supply $HY_2 (= OY_2)$. In this case, firms will be unable to sell the entire amount produced ($HY_2 > JY_2$) and hence their stock of inventories will increase (i.e., there will be unplanned inventory accumulation by firms). This would induce firms to reduce production in the next period. This process will continue till the equilibrium output $EY_e (= OY_e)$ is reached, where demand and supply are equal and there are no more unanticipated changes in firms' stock of inventories.

From this discussion we see that output market equilibrium in the Keynesian model is stable. Any situation of disequilibrium tends to be self-correcting, inducing a movement back towards the equilibrium level of output.

Three-Sector Economy with Government

In equation (2.4) we derived equilibrium output in a three sector model, with households, firms and a government sector. Diagrammatically this can be shown in Fig. 2.4. As earlier, the OL line makes an angle of 45° with the horizontal axis along which we measure income (or output). Aggregate demand is measured along the vertical axis. Now, this has three components, C, I and G, where investment and government expenditure are autonomous.

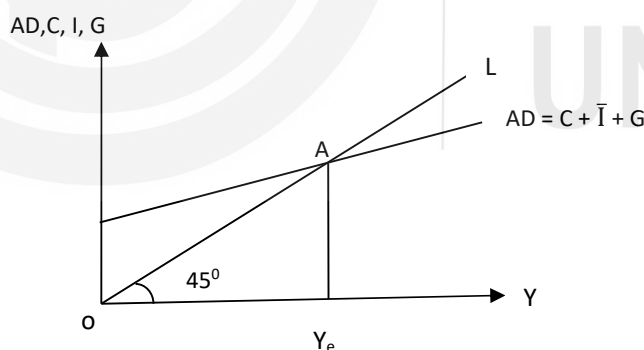


Fig. 2.4: Equilibrium Output in a Three-sector Economy

Equilibrium in the three-sector model occurs at the intersection of the AD curve and the OL line, where demand AY_e is equal to aggregate supply OY_e . You should see that equilibrium income level OY_e is unique and that this equilibrium is stable. At all income levels higher than OY_e there is excess supply, so that firms decrease production, inducing a change in output towards OY_e . On the other hand, at all income levels lower than OY_e , there is excess demand, inducing firms to increase production.

Note that the position of the AD curve is determined by the size of the autonomous components of aggregate demand (viz., I , G and autonomous consumption a). The higher the levels of these components, the higher would be the AD line and the higher would be the equilibrium output level (Y_e), implying that output is demand-determined in the Keynesian model.

2.3.2 Investment Multiplier

We will now examine the comparative static properties of the Keynesian model and explain the concept of the multiplier.

Autonomous Investment

Consider once again the two-sector model with only firms and households, such that equilibrium income is $Y_e = (a + \bar{I})/(1 - b)$. If, *ceteris paribus*, there is a change in the level of autonomous investment (ΔI), the associated change in income would be given by $\Delta Y = (1/(1 - b)) \times \Delta I$. Thus, $\frac{\Delta Y}{\Delta I} = (1/(1 - b))$. Here, we consider change in one component of autonomous demand at a time. This means when $\Delta I > 0$, we shall assume that $\Delta a = 0$ and so on. The term $(\frac{1}{1-b})$ is greater than one, since our assumption is that $(0 < b < 1)$. An implication of the above is that when investment increases by ΔI , the associated increase in income is greater than ΔI , as it is a multiple of ΔI . How is this possible? To see this, consider the several rounds of increments in expenditure that are triggered by a onetime increase in autonomous expenditure.

If we increase I by ΔI , income will increase by ΔI in the first round, which will induce a second round increase in consumption by $b\Delta I$ (since consumption is a function of income). Since output is demand-determined, this increase in consumption demand raises output and hence income increases further by $b\Delta I$. This induces a third round increase in consumption by $[b(b\Delta I) = b^2\Delta I]$ which triggers further increases in income ($b^2\Delta I$). In the fourth round, consumption demand increases by $[b(b^2\Delta I) = b^3\Delta I]$. The process continues with increase in consumption raising output, income and yet another round of consumption. Note that since $MPC < 1$ (i.e., $b < 1$), each subsequent round of increase in consumption is weaker than the previous one, so that the entire process dies down gradually.

Therefore, the total increase in income due to a onetime increase in autonomous investment is:

$$\Delta Y = (\Delta I + b\Delta I + b^2\Delta I + b^3\Delta I + \dots) = \Delta I (1 + b + b^2 + b^3 + \dots)$$

...(2.6) Since $0 < b < 1$, the expression in parentheses on the right hand side of (2.6) can be expressed as the sum of a convergent infinite geometric progression series, i.e., $1 + b + b^2 + b^3 + \dots = 1/(1 - b)$. Therefore, $\Delta Y = [1 / (1 - b)] \Delta I$, which shows that income increases by a multiple of the initial increase in autonomous investment and the multiplier is $(\frac{1}{1-b})$. This is shown diagrammatically in Fig. 2.5.